

Experiment 'p_4_1_e_sum_odd_digits' Results

June 5, 2026

Experiment outcome: SUCCESS

Bad responses: 0

Responses containing assume : 0

Responses containing expect : 0

Resolution attempts: 1

Hard fails (resolution): 0

Soft fails (resolution): 0

Verification attempts: 1

Problem Specification

Problem name: p_4_1_e_sum_odd_digits

Natural language statement: null

Method signature: p_4_1_e_sum_odd_digits(a: int) returns (sum: int)

Ensures

- `sum >= 0`
- `sum == s_o_d(a)`

Requires

- `a >= 0`

Functional Code Given

```
function s_o_d(a: int): int
requires a >= 0
{
  if a == 0 then 0
  else if a % 2 == 0 then s_o_d(a / 10)
  else a % 10 + s_o_d(a / 10)
}
```

GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

Program → GenAI

You are given the following task to perform in Dafny:

Write a method to satisfy the below requirements.

The signature should be:

`p_4_1_e_sum_odd_digits(a: int)` returns (`sum: int`)

The method should respect the following contract:

requires `a >= 0`, ensures `sum >= 0`, ensures `sum == s_o_d(a)`

The contract uses the following dafny code:

```
function s_o_d(a: int): int
requires a >= 0
{
  if a == 0 then 0
  else if a % 2 == 0 then s_o_d(a / 10)
  else a % 10 + s_o_d(a / 10)
}
```

These function(s) must not be used in your implementation of the method.

Produce and show only the Dafny body of this method, including the curly braces that
→ surround it. Do not show the signature nor contract. You must not use 'assume' nor
→ 'expect' anywhere in your code.

GenAI → Program

System fingerprint: null

ID: chatcmpl-DnLHIxa1Heilt6LVabUUnUuj9sh1d

Model: gpt-5.5-2026-04-23

Created at: 1780652236

Finish reason: stop

Usage: 446 tokens in, and 618 tokens out

```
{
  if a == 0 {
    sum := 0;
  } else if a % 2 == 0 {
    sum := p_4_1_e_sum_odd_digits(a / 10);
  } else {
    var rest := p_4_1_e_sum_odd_digits(a / 10);
    sum := a % 10 + rest;
  }
}
```

```
}
```

Final Program

```
function s_o_d(a: int): int
requires a >= 0
{
  if a == 0 then 0
  else if a % 2 == 0 then s_o_d(a / 10)
  else a % 10 + s_o_d(a / 10)
}

method p_4_1_e_sum_odd_digits(a: int) returns (sum: int)
  requires a >= 0
  ensures sum >= 0
  ensures sum == s_o_d(a)
{
  if a == 0 {
    sum := 0;
  } else if a % 2 == 0 {
    sum := p_4_1_e_sum_odd_digits(a / 10);
  } else {
    var rest := p_4_1_e_sum_odd_digits(a / 10);
    sum := a % 10 + rest;
  }
}
```

Total Token Usage

Input tokens: 446

Output tokens: 618

Reasoning tokens: 512

Sum of 'total tokens': 1064

Experiment Timings

Overall Experiment started at 1780652236111, ended at 1780652260060, lasting 23949ms (23.95 seconds)

Iteration #1 started at 1780652236234, ended at 1780652260060, lasting 23826ms (23.83 seconds)